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ACAT1/ACAT2 as biomarkers for idiopathic pulmonary fibrosis: insights from transcriptomics analysis and machine learning

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Title: ACAT1/ACAT2 as biomarkers for idiopathic pulmonary fibrosis: insights from transcriptomics analysis and machine learning

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Abstract

Background: Idiopathic pulmonary fibrosis (IPF) is a devastating interstitial lung disease characterized by dysregulated inflammation and progressive lung scarring, ultimately culminating in mortality due to respiratory complications. Metabolic reprogramming in IPF progression has been extensively studied, yet the role of tryptophan metabolism remains poorly understood. This study aimed to identify key tryptophan metabolism-related genes (TMGs) as diagnostic biomarkers for idiopathic pulmonary fibrosis (IPF) and elucidate their cell-type-specific roles in IPF pathogenesis.

Methods: Public bulk RNA sequencing of human IPF datasets and single-cell RNA-seq (scRNA-seq) data were analyzed to assess the expression level of TMGs. Machine learning algorithms (Boruta, LASSO, SVM-RFE, XGBoost, random forest) identified critical TMGs. Validation included immunohistochemistry, qRT-PCR, Western blot, and immunofluorescence in bleomycin-induced mouse models and human IPF lung tissues. Further scRNA-seq analysis characterized cell-type-specific TMG expression and metabolic activity.

Results: *ACAT1* and *ACAT2* were identified as core TMGs significantly downregulated in IPF, as identified by transcriptomic profiling and machine learning. ScRNA-seq revealed reduced AT2 cell abundance and suppressed TMG activity in IPF. Correlation analyses linked ACAT1/ACAT2 to metabolic pathways (e.g., amino acid metabolism) and fibrosis-related processes. Mouse and human validation confirmed decreased ACAT1/ACAT2 expression in fibrotic lungs, correlating with AT2 cell dysfunction. Inhibition of ACAT1/ACAT2 using Avasimibe significantly aggravated fibrosis.

Conclusion: ACAT1 and ACAT2 are novel diagnostic biomarkers and potential therapeutic targets for IPF, with their suppression in AT2 cells reflecting metabolic dysregulation central to fibrosis progression. This study underscores the therapeutic potential of targeting AT2 cell metabolic

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homeostasis to mitigate IPF. Multi-omics and machine learning approaches provide a robust framework for identifying metabolic biomarkers in complex diseases.

Keywords: idiopathic pulmonary fibrosis; tryptophan metabolism; ACAT1; ACAT2; machine learning.

Introduction

Idiopathic pulmonary fibrosis (IPF) is a progressive and fatal interstitial lung disease characterized by abnormal extracellular matrix deposition, alveolar epithelial cell dysfunction, and irreversible structural distortion of the lung parenchyma[1]. Although advances have been made in understanding its pathogenesis, the molecular drivers of IPF remain incompletely defined, and current therapeutic strategies only modestly slow disease progression[2].

Emerging evidence highlights metabolic reprogramming as a critical contributor to fibrotic processes[3]. Transcriptomic studies suggest widespread metabolic alterations in IPF, with localized dysregulation of metabolic pathways implicated in pro-fibrotic mechanisms such as immune imbalance, oxidative stress, and epithelial injury[4]. Metabolic reprogramming or dysfunction in diverse cell types—including fibroblasts, epithelial cells, immune cells, and endothelial cells—has been linked to fibrotic pathogenesis[5]. For instance, epithelial cells and macrophages in IPF exhibit a "Warburg effect," favoring glycolysis over oxidative phosphorylation even under normoxic conditions[6,7].

The roles of small-molecule metabolism in pulmonary fibrosis are increasingly recognized[8]. Lipid metabolism, for example, may influence fibrosis through enhanced fatty acid synthesis and reduced fatty acid oxidation[9], while amino acid metabolism governs cell survival, energy supply,

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biosynthetic precursors, and tissue homeostasis[10]. Studies suggest that altered amino acid metabolism during fibrosis primarily involves non-essential amino acids, with increased catabolism promoting collagen synthesis and disease progression[11].

However, the expression patterns, diagnostic potential, and cell type specificity of tryptophan metabolism-related genes (TMGs) in IPF pathogenesis remain poorly understood. Tryptophan metabolism regulates diverse biological processes—including immune tolerance, mitochondrial energy metabolism, and redox homeostasis—through metabolites such as kynurenine and serotonin[12]. Dysregulation of this pathway is implicated in chronic inflammatory and fibrotic diseases[13,14], yet its role in IPF remains elusive. Notably, tryptophan metabolites exhibit extensive crosstalk with other metabolic pathways, such as amino acid and lipid metabolism[15,16]. Despite this interconnectedness, systematically identifying key TMGs with diagnostic and mechanistic relevance in IPF remains a critical challenge. Furthermore, cellular diversity and heterogeneity in lung tissues complicate the interpretation of bulk transcriptomic data, necessitating high-resolution approaches to dissect cell-type-specific metabolic dysfunction.

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In this study, we integrate bulk RNA sequencing, machine learning-driven feature selection, and single-cell transcriptomic profiling to identify TMGs as diagnostic biomarkers and uncover core cell populations exhibiting pathological metabolic disruptions during IPF progression. By analyzing IPF-related transcriptomic datasets, applying machine learning algorithms, and validating findings in animal models and patient-derived pathological samples, we identified ACAT1 and ACAT2 as core TMGs significantly downregulated in IPF. These genes were predominantly expressed in alveolar type 2 (AT2) epithelial cells—the primary cellular hub of tryptophan metabolism in the lung. In IPF, AT2 cells exhibited metabolic suppression marked by reduced ACAT1 and ACAT2

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expression, accompanied by a decline in cell abundance.

This study provides the first comprehensive evidence that ACAT1 and ACAT2 may serve as metabolic hubs in IPF pathogenesis, with their reduced expression offering a novel diagnostic indicator. Our work establishes a framework for identifying metabolic biomarkers in complex diseases and underscores the therapeutic potential of restoring metabolic homeostasis in AT2 cells during fibrotic lung repair.

Materials and Methods

Mouse Models

8-week-old male C57BL/6J (wild type) mice were obtained from the Shanghai Animal Center and randomized into experimental groups. Mice were housed under an ambient temperature of 24 ± 2 °C, circulating air, constant humidity of 50 ± 10% and a 12 h:12 h light/dark cycle. The animal sample size for each experiment was selected based on previous well-characterized studies (mostly 6 per group). The specific number of animals in each group is specified in the figure legends. All experiments using mice were approved by the Ethics Committees of Shanghai Pulmonary Hospital. We complied with all relevant ethical regulations and used approved study protocols.

An experimental IPF model was induced by intratracheally instilled bleomycin in 8-week-old male mice. The mice were intratracheally instilled with 50μl saline or bleomycin (BLM) at a dose of 5.0 mg/kg body weight, diluted in 50μl saline, to set up the control or IPF model on day 0. Lung-specific *Acat2* knockdown mice were generated by injecting Ad-sh*Acat2* (1.0×109 U/mice) (Viraltherapy Technologies, China) intratracheally at Day 0 and Day 10. At day 28, mice were sacrificed. The left lung was fixed in 10% buffered formalin for 48h for histological analysis and the right lung was

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collected for western blot and PCR.

Mouse lung function test

Lung function parameters such as dynamic compliance and respiratory resistance were determined using the Pulmonary Function Test System (Buxco Research Systems, USA) according to the manufacturer's protocol. Briefly, mice were first anesthetized with Avertin solution (T48402, Sigma-Aldrich), and transferred into the body chamber of the detecting system.

Human subjects

Human IPF lung tissue was obtained from patients undergoing surgery for organ transplantation in Shanghai Pulmonary Hospital. Control samples were obtained from non-IPF histology lung samples from patients with lung cancer in the hospital. IPF was diagnosed by chest high-resolution CT images and other examinations according to the respective diagnostic criteria for this disease[17]. Informed written consent was obtained from all participants. The research has been approved by the Ethics Committees of Shanghai Pulmonary Hospital. All human sample studies were performed in compliance with the official ethical regulations.

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qRT-PCR

Total RNA from cells or tissues was extracted using TRIzol reagent (R401, Vazyme) for qPCR, which was subsequently reverse-transcribed into cDNA with HiScript II Q RT SuperMix (R222-01, Vazyme). Primers used for ACAT1 and ACAT2 were both purchased from Beyotime Company (QM51322S and QM51330S, Beyotime). An Applied Biosystems 7500 device with ChamQ SYBR qPCR Master Mix (Q331-02, Vazyme) was used to conduct quantitative real-time polymerase chain reaction (qRT-PCR). Expression data were normalized to β-actin mRNA expression and fold-change was calculated using $2^{-\Delta\Delta Ct}$ approach.

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Western Blots

The collected proteins were separated via SDS-PAGE procedure and subsequently electrotransferred to polyvinylidene fluoride (PVDF) membranes. After 1 hour block at room temperature, membranes were reacted with the corresponding primary antibodies, of which the details are ACAT1 (1:1000, 16215-1-AP, Proteintech), ACAT2 (1:1000, 14755-AP, Proteintech) and GAPDH (1:1000, T0004, Abmart), overnight at 4°C. Incubation was conducted consequently with HRP-linked rabbit/mouse IgG antibody (1:3000,7074S/ 7076S, Cell Signaling Technology) for 1 hour.

Histopathology of mice lung sections

Lung tissue sections were prepared for histological examination using both Hematoxylin and Eosin (HE) and Masson's Trichrome staining. For HE staining, tissues were fixed in 10% neutral buffered formalin, paraffin-embedded, sectioned at 4 μm, and stained with hematoxylin followed by eosin to visualize nuclei and cytoplasm, respectively. Masson's Trichrome staining was employed to assess collagen fiber deposition, with tissues first stained with Wiegert's hematoxylin, then with Biebrich scarlet-acid fuchsin to highlight collagen, and finally with aniline blue to differentiate muscle and connective tissues. Each staining step was followed by appropriate washing, differentiation, and dehydration processes before mounting with a synthetic medium. Slides were examined under a light microscope to evaluate tissue morphology and fibrosis.

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Immunofluorescence and Immunohistochemical staining

The lung tissues from mice were insufflated with a mix of 1:1 Water/OCT and quickly frozen at -80°C. Slides from those lungs were fixed in methanol for 15 min before being blocked in 5% FBS. Primary antibodies for ACAT1, ACAT2 and SFTPC were incubated overnight at 4°C. Goat anti-

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rabbit and goat anti-mouse conjugated with Alexa Fluor 594 or Alexa Fluor 488 (Molecular Probe, InVitrogen, Cergy Pontoise, France) were used as secondary antibodies at a dilution of 1: 2000.

Lung tissues were dehydrated and embedded in paraffin. For histological examination, 4-μm-thick tissue sections on slides were treated with 1.4% H2O2–methanol for 30 min to block endogenous peroxidase. Then, nonspecific binding was blocked with 1.5% normal serum, and the slides were incubated with rabbit anti-ACAT1 and anti-ACAT2 polyclonal antibodies (1:500; 16215-1-AP, Proteintech and 1:500; 14755-AP, Proteintech). The sections were incubated with SABC-HRP Kit (P0603, Beyotime). The color reaction was developed by incubation with a 3,3′-diaminobenzidine liquid substrate system (D3939, Sigma-Aldrich). The slides were counterstained with Harris's hematoxylin for 1 min after immunohistochemical staining.

Ashcroft scoring

Ashcroft scoring system was used to estimate the severity of pulmonary fibrosis. Criteria for grading lung fibrosis are as follows. 0 point: normal lung; 1 point: minimal fibrous thickening of alveolar or bronchiolar walls; 3 points: moderate thickening of walls without obvious damage to lung architecture; 5 points: Increased fibrosis with definite damage to lung structure and formation of fibrous bands or small fibrous masses; 7 points: severe distortion of structure and large fibrous areas; "honeycomb lung" is placed in this category; 8 points: total fibrous obliteration of the field. Scores of 2, 4, and 6 represent intermediate stages between the respective points.

Transcriptome Datasets Acquisition and Analyses

"Pulmonary fibrosis" and "Lung fibrosis" were used as keywords to search for related RNA-seq datasets in the Gene Expression Omnibus (GEO) database (https://www.ncbi.nlm.nih.gov/geo/). Two datasets of samples of human lung tissues were selected (GSE231693, GSE52463). The

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GSE231693 dataset was based on the GPL20301 platform (Illumina HiSeq 4000 [Homo sapiens]). The GSE52463 dataset was based on the GPL11154 (Illumina HiSeq 2000 [Homo sapiens]). Both above included RNA-seq samples from the lung tissues of IPF patients and healthy controls. The integrated expression matrix could be seen in **Supplementary material 5**. Public scRNA-seq data were acquired from gene expression matrices of GSE122960[18] and GSE190889[19] on the GEO database (four IPF samples and control samples were selected from GSE122960, three IPF samples and control samples were selected from GSE190889).

The two datasets were integrated into a unified expression matrix in R (version 4.3.1). To eliminate technical variations between different sequencing platforms, batch effect correction was performed using the removeBatchEffect function from the limma package. To ensure the integrity of the integrated data, we first performed gene ID conversion using the org.Hs.eg.db package, mapping probe or transcript IDs to official Gene Symbols. For genes with multiple probes or transcripts, the median expression value was retained. To eliminate technical variations and systemic biases between different sequencing platforms and batches, the removeBatchEffect function from the limma package was applied to the normalized expression matrix. Differentially expressed genes (DEGs) between IPF patients and healthy controls were identified using the DESeq2 package based on the integrated count matrix. Genes that meet statistical significance threshold of adj. P-value < 0.05 and |log2FoldChange| > 1.5 were defined as significant DEGs for subsequent analysis[20]. Volcano plots were made by the package ggplot2.

To identify IPF-associated gene modules, WGCNA was performed on the batch-corrected matrix. A soft-thresholding power (β) was selected to ensure scale-free topology ($R^2 > 0.85$). We calculated Module-Trait Relationships (MTRs) to identify the key module most significantly

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correlated with IPF. Genes with high module membership (MM > 0.8) and gene significance (GS > 0.2) were prioritized as candidate biomarkers.

Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses were performed by the R package cluster profiler. Afterwards, the org.Hs.eg.db (version 3.4.0) and enrichplot R packages were used to realize the visualization of enrichment analysis results.

ScRNA-seq Analysis

All scRNA-seq data were first processed using the Seurat R package version 2.3.4. We treated three datasets in the same steps as follows. All Seurat objects were combined into a merged dataset, and a percentage of mitochondrial genes was calculated for each sample in the merged object. For quality control, cells containing fewer than 500 identified genes or more than 10% of reads arising from mitochondrial genes were removed. SCTransform with default parameters was used to normalize and scale the data. Dimensionality reduction was performed on the top 2000 most variable genes using PCA. Clustree R package was used to determine the best resolution when clustering. Different groups were manually annotated with known cell type markers reported by Reyfman, et al[18]. Violin plots and UMAP plots overlaid with gene expression level were generated using Seurat. Differentially expressed genes of each cluster were identified using FindAllMarkers function of Seurat. R package ggplot2 was used to prettify default figures created using Seurat.

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Machine Learning

Five independent machine learning algorithms were used to identify optimal key genes in idiopathic pulmonary fibrosis (IPF): Boruta, LASSO, SVM-RFE, XGBoost, and random forest. These algorithms were chosen for their complementary strengths in feature selection, model optimization, and bias reduction, which enhanced the robustness and accuracy of identifying IPF-specific genes[21].

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The Boruta algorithm was selected for its rigorous identification of all relevant features associated with the target variable, ensuring retention of only the most important genes. It iteratively compared original features with randomly permuted shadow attributes, retaining only those most relevant to IPF while eliminating noise. LASSO, implemented using the “glmnet” package, introduced a regularization term to shrink coefficients, selecting the most predictive features while eliminating irrelevant or redundant genes. SVM-RFE, a feature elimination method based on support vector machines, iteratively removed the least important features to refine the set of key genes contributing most to classification accuracy. XGBoost and random forest were included to improve prediction reliability. XGBoost, based on the gradient boosting framework, identified the optimal predictive model through multiple iterations. Random forest built multiple decision trees and averaged their outcomes to rank feature importance, allowing selection of the top 7 diagnostic gene candidates. The overlapping genes identified by these five algorithms were selected as the core TMGs in IPF and visualized using a Venn diagram. The model's performance was evaluated using 10-fold cross-validation within the training set. For external validation (GSE150910), we strictly applied the pre-trained model (without re-training) to the log2-transformed and Z-score normalized validation data. ROC analysis was performed using the pROC package to assess the model's diagnostic accuracy.

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Image Analyses

Screened pictures of stained slides were examined in professional pathology analysis software CaseViewer (version 2.4.0., 3DHISTECH). Quantification of fluorescence strength or protein expression level was completed in ImageJ.

Quantification and Statistical Analysis

All experiments were performed technically three times independently to improve the data

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reliability. The one-way analysis of variance (ANOVA) for multiple group comparisons was utilized to assess the statistical significance of differences, and statistical differences between 2 groups were computed by an unpaired two-tailed t test. Experimental statistical analyses and visualization were performed in Graphpad Prism 9.0.0 for Windows (GraphPad, San Diego, CA, U.S.A). At least four biological replicates were performed for all experiments.

Results

Bulk RNA-seq analysis of IPF samples

We first removed the batch effect to integrate bulk RNA-seq data acquired from GSE52463 and GSE231693 using the function "removeBatchEffect" in R package limma (**Fig. 1A**). To demonstrate the expression level of tryptophan metabolism genes (TMGs, see in **Supplementary material 4**), a volcano plot coloring differential expressed genes (DEGs) was generated. Several genes both being in TMGs and DEGs (Down-regulated: *ACAT2*, *KMO*, *CYP1A2*; Up-regulated: *HAAO*, *AOC*, *ALDH1B1*) were pointed out in the plot (**Fig. 1B**). We next conducted enrichment analysis of DEGs to build a macro atlas of biological changes from healthy control samples to IPF samples. GO enrichment analysis showed several terms with positive z-scores like cilium movement, immunoglobin complex and antigen binding. In the category of cellular component (CC), terms like collagen-containing extracellular matrix and collagen trimer demonstrated significant fibrosis features of IPF (**Fig. S1A**). KEGG enrichment showed similar results (**Fig. S1B**). However, it's noticeable that quite a lot of metabolism pathways, especially those of synthesis like terpenoid backbone synthesis, steroid biosynthesis and biosynthesis of unsaturated fatty acid were inhibited in IPF status, indicating that metabolism dysfunction played an important role in IPF pathogenesis.

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To find out whether part of a series of metabolism pathways is injured, we performed weighted gene co-expression network analysis (WGCNA) of the integrated expression matrix. Both scale free topology model fit and mean connectivity preferred five as the best soft threshold for analysis (**Fig. S1C**). Cluster dendrogram displayed 12 different expression modules, among which the cyan module had the most negative relationship with IPF (**Fig. S1D and E**). Genes in cyan module (see in **Supplementary material 7**) were mostly enriched into metabolism terms belonging to groups named like "amino cellular small molecular" and "acyl-CoA fatty-acyl-CoA thioester derivative" (**Fig. 1C**). Moreover, the cyan module genes shared three genes with TMGs: *ACAT1*, *ACAT2*, *GCDH* (**Fig. 1D**). Heatmaps further defined the down regulation of these genes (**Fig. 1E**). These findings again pointed out that blocked metabolism, such as amino acid or fatty acid metabolism, may be responsible for the pathogenesis of IPF.

Machine learning-selected TMGs demonstrated diagnostic value in IPF

Bulk-seq transcriptomic analysis revealed the relationship between TMGs and IPF. To screen out genes with the max diagnostic value in the training set, five machine learning algorithms, including the LASSO algorithm (**Fig. 2A**), the Boruta algorithm (**Fig. 2B**), the SVM-RFE algorithm (**Fig. 2C**), the random forest algorithm (**Fig. 2D**), and the XGBoost forest model (**Fig. 2E**) were employed. Through intersection analysis of the genes co-selected by five algorithms, six key feature genes were identified: *ACAT1*, *ACAT2*, *ALDH1B1*, *AOC1*, *CYP1A2*, *HAAO* (**Fig. 2F**). However, as *CYP1A2* rarely expressed in tissues but liver, we excluded it and built a diagnosis model using the other five selected genes. Receiver operating characteristic (ROC) curve of these genes revealed their individual diagnostic capabilities (**Fig. S2A-E**). Diagnostic ROC curve showed that the area under the curve (AUC) for the combined model was 0.860. ROC curve of these genes revealed their

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individual diagnostic capabilities (**Fig. 2G**). Moreover, the specific diagnostic capabilities of different gene markers for IPF were further displayed through the nomogram (**Fig. 2H**). Decision Curve Analysis (DCA) indicated that IPF patients could achieve significant net benefits (**Fig. 2I**), and calibration curves demonstrated a tiny difference between the actual risk of IPF and the predicted risk, highlighting the high accuracy of the model (**Fig. 2J**).

To validate the model accuracy, we used a dataset of GSE150910 (103 samples of control and 103 samples of IPF) to build our validation set. The confusion matrix based on the validation set showed a robust prediction power of the model (**Fig. 2K**). However, when testing the individual genes' diagnostic capabilities, only *ACAT1*, *ACAT2*, and *AOC1* performed well, corresponding to the differential expression between IPF and control samples shown by violin plots (**Fig. 2L, S3A-E**).

The scRNA-seq profiling of IPF revealed a decreased number of AT2 cells

To discover and further validate the status of genes screened out by machine learning at the single-cell level, we collected human IPF and control sample data from GSE122960 and GSE190889. Then a total of 14 samples were chosen for further analysis, comprising 7 from the control group and 7 from the IPF group. To address potential batch effects, a correction method was applied across all samples. The results indicated that the overall distribution remained relatively stable, with minimal sensitivity to batch effects (**Fig. S4A**). A uniform manifold approximation and projection (UMAP) plot demonstrated cell distribution of different samples. Subsequently, the Seurat pipeline analysis was employed, which grouped all cells into 16 distinct subpopulations (**Fig. S4B**). 12 cell types, including B cells, T cells, fibroblasts, and others were identified through the use of specific marker genes and annotated in UMAP plot (**Fig. S4C, D**). The expression patterns of characteristic marker genes associated with 12 cell subsets were also visualized in UMAP plots (**Fig. S4E**). To elucidate

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the differences between the IPF and control groups, a comparison of the proportions of different single cell types was performed. Surprisingly, the proportion of alveolar type 2 (AT2) cells was significantly higher in the control group (**Fig. S4F**). This suggests that the absence of AT2 cells may play a crucial role in the pathogenesis of IPF.

TMGs are predominantly expressed in AT2 cells

The role of the tryptophan metabolic pathway in the progression of IPF remains to be elucidated, and its metabolic status in IPF patients is currently unknown. In this study, we employed four algorithms—AUCell, UCell, singscore, and ssGSEA to assess the status of tryptophan metabolism in IPF at the single-cell level. The expression levels of TMGs across different cell types were evaluated by calculating the average scores derived from the aforementioned algorithms. The results revealed significant heterogeneity in the activity of TMGs across different disease states and cell types. Specifically, during IPF, the activity scores of TMGs were significantly reduced (**Fig. 3A**). These genes exhibited higher (statistically significant) activity in AT2 cells and dendritic cells but were relatively suppressed in basal cells, ciliated cells, club cells, endothelial cells, monocytes, and T cells (**Fig. S5A**). Ridge plots and violin plots demonstrated TMGs scores in all cell types individually (**Fig. S5B and C**). Further visualization using a UMAP plot to display the distribution of TMG scores under the four scoring methods indicated that these genes were predominantly expressed in AT2 cells and macrophages (**Fig. 3B**). By intersecting the tryptophan metabolism gene set with the differentially expressed genes in AT2 cells and macrophages between the IPF and control groups, we identified differentially expressed TMGs at the single-cell level: *ACAT1*, *ACAT2* and *ALDH3A2* showed a significant down regulation in AT2 cells of IPF samples, and IL4I1 was up-regulated in macrophages when in IPF (**Fig. 3C-F**). Linking to our former results, *ACAT1* and

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ACAT2 showed the greatest potential role in IPF pathogenesis. In that, we defined these two genes in TMGs, *ACAT1* and *ACAT2*, as the key genes in IPF.

Correlation analysis of key genes reveals cellular biological functional changes in IPF

Due to complex communications in the ocean of genes, we did a correlation analysis from scRNA-seq validated TMGs, especially *ACAT1* and *ACAT2* (*ALDH3A2* was also included in the analysis), to have a better understanding of the impact of inhibited expression level of defined key genes in AT2 cells. Heatmap demonstrated that genes like *PLA2G1B*, *SLC25A5* and *LDHB*, which respectively involved in phospholipid degradation, amino acid transportation and lactate generation, showed a positive relationship with *ACAT1* (down-regulated in IPF samples) (**Fig. 4A**). Same pattern occurred in *ACAT2* related genes (**Fig. 4B**). Genes positively related with *ALDH3A2* involved in cell adhesion and signaling transduction (*CD24*, *CSTE*, *CHST3*) and anti-inflammation (*CP*, *SERPIND1*, *MMP9*) (**Fig. 4C**). However, negatively correlated genes with these three key genes showed no significant specificity in biological function (**Fig. 4D-F**). To better figure out the macroscopic function of these genes related to *ACAT1*, *ACAT2* and *ALDH3A2*, we further did enrichment analysis. As the similar expression pattern of the three key genes, we first collect the union sets of their positive and negative related genes for later use. In GO enrichment analysis of positive related genes, enriched terms were mainly about small molecule metabolism, mitochondrial formation and oxidoreductase activity, indicating occurrence of inhibited matter and energy metabolism in IPF AT2 cells (**Fig. 4G**). KEGG enrichment of these genes also confirmed the fact, as it displayed terms like “pyruvate metabolism”, “valine, leucine, and isoleucine degradation”, “terpenoid backbone biosynthesis” and “tryptophan metabolism” (**Fig. 4I**). In GO enrichment analysis of negative related genes, which were up-regulated in IPF, enriched terms were mainly

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related with cell adhesion, extracellular matrix formation and activities involves in inflammation, corresponding to the features of IPF (**Fig. 4H**). KEGG analysis showed similar results (**Fig. 4J**). Taken together, the inhibition of key TMGs, especially *ACAT1* and *ACAT2*, indicated a much more severe metabolism injury in AT2 cells in IPF. It not only underlined the importance of *ACAT1* and *ACAT2* in IPF, but also dropped a hint that metabolic function of AT2 cells may play a key role in IPF progression. Our results here revealed downregulated *ACAT1* and *ACAT2* expression level could be defined as risk factor or diagnostic tool for IPF.

Mouse models and clinical IPF samples exhibited downregulation of ACAT1 and ACAT2

Former results noticed us the possible expression pattern of ACAT1 and ACAT2 in IPF environment. Then we first detected the expression of ACAT1 and ACAT2 in mouse lung slices (from Bleomycin induced fibrosis group and control group) by immunohistochemical (IHC) staining. Signals both from ACAT1 and ACAT2 were weakened in samples of IPF mouse. Interestingly, main ACAT1 and ACAT2 positive cells were found to be AT2 cells morphologically (**Fig. 5A, B**). Next, prosurfactant protein C (SFTPC), a widely used marker of AT2 cells, was co-stained with ACAT1 and ACAT2 under immunofluorescence (IF) procedures in mouse lung slices. Similar to the results of bioinformatic analysis and IHC staining, it's obvious that the expression of ACAT1 and ACAT2 were inhibited in IPF AT2 cells, compared to the control group (**Fig. S6A, B**). Homogenates from mouse lungs were used to quantify the expression of ACAT1 and ACAT2 at transcriptional and translational level. Both q-PCR and Western blot results revealed that the expression of ACAT1 and ACAT2 was significantly down-regulated in IPF mouse lung tissues (**Fig. 5C, D**). After confirming the expression pattern in mouse model, we similarly detected the expression of ACAT1 and ACAT2 in human lung biopsy samples from IPF patients and non-IPF patients. Through IF staining, it's

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clear that AT2 cells expressed ACAT1 and ACAT2 dominantly (**Fig. 5E**). Same with the phenomenon in IPF mice model, AT2 cells in IPF patients demonstrated lower ACAT1 and ACAT2 expression, indicating a metabolism dysregulation mediated by ACAT1 and ACAT2 occurred in IPF (**Fig. 5F, G**). Interestingly, ACAT2 showed a better co-expression model with SFTPC in both human and mouse tissue slices. All in all, these results again enhanced the diagnostic value of ACAT1 and ACAT2 expression level for IPF. Moreover, the results indicated that it's of great importance for IPF control to maintain the stable metabolism function of AT2 cells, especially metabolisms mediated by ACAT2.

Knockdown of ACAT2 exacerbated pulmonary fibrosis

Previous results have confirmed that the expression of ACAT1 and ACAT2 is significantly downregulated during pulmonary fibrosis, suggesting that their mediated metabolic processes may be impaired. However, whether the loss of ACAT function directly exacerbates fibrosis remains to be elucidated. So we employed adenovirus-mediated knockdown of ACAT2 (Adv-shACAT2), administered via intratracheal nebulization to both pulmonary fibrosis model mice and control group mice. H&E staining revealed that the alveolar septa were partially thickened in mice treated with Adv-shACAT2 alone, potentially indicative of early fibrotic changes. Mice subjected to both BLM and Adv-shACAT2 exhibited more severe fibrosis than the BLM-induced group; histological sections showed marked alveolar structural disruption and significant deposition of hemosiderin within macrophages (**Fig 6A**). Masson's trichrome staining further demonstrated increased collagen deposition following ACAT2 knockdown (**Fig 6B**). Correspondingly, the Ashcroft scores for these representative sections were significantly higher in the Adv-shACAT2-treated mice (**Fig 6C**).

Notably, while mice treated solely with Adv-shACAT2 did not show significant pulmonary function

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impairment compared to the control group, the knockdown of ACAT2 on top of the BLM-induced model effectively worsened pulmonary function (**Fig 6D, E**). Following histological analysis, total RNA was extracted from mouse lung homogenates for downstream quantification. The significant upregulation of a series of fibrosis-related genes confirmed that ACAT2 deficiency promotes the progression of fibrosis (**Fig 6F**). In summary, the expression of ACAT1/ACAT2 is reduced in IPF lung tissue. The fact that knocking down ACAT2 worsens pulmonary fibrosis indicates that their expression levels can serve as effective biomarkers and potential metabolic therapeutic targets, providing a new direction for future research.

Discussion

Consistent with our research objectives, our findings demonstrate significant downregulation of ACAT1 and ACAT2 during pulmonary fibrosis, predominantly localized to alveolar type II (AT2) cells, with ACAT2 showing the most pronounced suppression. Notably, we observed a concurrent reduction in AT2 cell abundance in fibrotic lungs, further attenuating the metabolic intensity mediated by ACAT1 and ACAT2. This AT2-centric metabolic reprogramming likely represents a critical pathogenic mechanism in fibrosis, while the expression levels of ACAT1 and ACAT2 may serve as novel biomarkers for IPF diagnosis.

Recent studies have increasingly highlighted the central role of metabolic reprogramming in pulmonary fibrosis, and our results align with emerging literature[22]. For instance, alterations in non-essential amino acid metabolism, such as glutamine[23], glycine[24], and arginine[25], have been linked to pro-fibrotic cellular phenotypes in IPF. Specifically, TGF-β1 promotes glutaminolysis via upregulation of glutaminase (GLS) to fuel myofibroblast differentiation[26], while arginine

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metabolites like asymmetric dimethylarginine (ADMA) enhance collagen synthesis by activating arginase-1 (Arg-1)[27]. Although ACAT1 and ACAT2 are not part of the rate-limiting kynurenine pathway, their identification within the TMG framework reflects the terminal integration of tryptophan catabolism. Tryptophan is a ketogenic amino acid; its degradation eventually generates acetoacetyl-CoA, which must be processed by ACAT1/2 to enter the TCA cycle for ATP production. Therefore, the downregulation of ACAT1/2 in IPF may represent a 'metabolic bottleneck' that prevents the efficient utilization of tryptophan-derived carbons, potentially leading to the accumulation of metabolic intermediates that fuel pro-fibrotic signaling or exacerbate oxidative stress in AT2 cells. While tryptophan derivatives such as 5-methoxytryptophan have been reported to attenuate fibrosis by inhibiting TGF-β/SMAD3 signaling[28], and serotonin exacerbates fibrosis via inflammation and oxidative stress[29], their direct roles in IPF pathogenesis require further validation. McGowan et al. previously implicated lipid metabolic dysregulation in fibroblasts as a driver of fibrosis[30]. Although ACAT1/ACAT2 were identified within the TMG framework here, their established roles in lipid and cholesterol metabolism expand upon the scope of metabolic dysregulation in IPF. ACAT1 (acetyl-CoA acetyltransferase 1) and ACAT2 (acetyl-CoA acetyltransferase 2), members of the thiolase family, catalyze the conversion of acetyl-CoA to acetoacetyl-CoA, playing pivotal roles in cholesterol synthesis and ketogenesis[31]. ACAT1, localized to mitochondria, governs ketone body production (particularly under energy stress) and branched-chain amino acid metabolism, whereas ACAT2, enriched in the endoplasmic reticulum, primarily regulates cholesterol ester synthesis and lipid storage in the liver and intestines[32]. Our study revealed significant expression of both enzymes in AT2 cells, with marked suppression in IPF. While lipid metabolism in IPF has garnered attention—for example, aberrant fatty acid/cholesterol synthesis

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and lipid-derived mediators (prostaglandins and leukotrienes) driving pro-fibrotic signaling[33,34], the specific mechanisms underlying ACAT1/ACAT2-mediated metabolic suppression remain underexplored, warranting further mechanistic studies.

AT2 cells occupy a central role in IPF pathogenesis, where their dysfunction reflects a collapse of the metabolic-regenerative axis[35]. As key effectors of alveolar repair, AT2 cells maintain homeostasis via surfactant protein secretion (e.g., SP-A, SP-C) and replenish damaged alveolar type I cells[36]. In IPF, AT2 cells undergo pathological remodeling[37]: recent research reveals impaired regenerative capacity, characterized by upregulation of cell cycle arrest genes like p21 or CDKN1A and activation of endoplasmic reticulum stress markers such as DDIT3[38,39]. Furthermore, AT2-specific metabolic reprogramming exacerbates oxidative damage and paracrine activation of fibroblasts via pro-fibrotic mediators (TGF-β and PDGF)[40]. Our findings align with prior work positioning AT2 cells as hubs of a metabolic-epigenetic-fibrotic network in IPF. The observed decline in AT2 cell abundance and suppression of ACAT1/ACAT2-mediated metabolism in IPF tissues further underscore their pivotal role in disease progression.

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While our study robustly demonstrates AT2-centric metabolic reprogramming via ACAT1/ACAT2 downregulation in IPF, several limitations still remain. We established a strong correlation between reduced ACAT1/ACAT2 expression and fibrosis, and provided initial evidence for causality. However, the specific mechanism between pulmonary fibrosis and ACAT1/ACAT2 function has not been discussed in this study. Prior works in IPF models identified elevated cholesterol levels in bronchoalveolar lavage fluid and reduced expression of cholesterol efflux genes in macrophages and AT2 cells. We hypothesize that the suppression of ACAT1/ACAT2 disrupts the mitochondrial acetyl-CoA homeostasis in AT2 cells. Acetyl-CoA is not only a primary fuel for the TCA cycle but

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also a key substrate for protein acetylation, a major post-translational modification that regulates autophagy, ER stress, and DNA repair. The reduction of ACAT-mediated thiolase activity may lead to energy failure and dysregulated acetylation of protective proteins in AT2 cells, thereby impairing their regenerative capacity and fostering a pro-fibrotic microenvironment. While prior studies noted lipid dysregulation in IPF, our results pivot the focus toward the metabolic-energetic axis governed by acetoacetyl-CoA processing. Additionally, the crosstalk between metabolically reprogrammed AT2 cells and neighboring fibroblasts or macrophages remains unexplored. Subsequent studies should employ cell-cell communication analyses and co-culture systems to dissect these interactions. In summary, this multi-omics and machine learning-driven study identifies ACAT1 and ACAT2, which are downstream TMG candidates, as significantly suppressed genes in IPF, with AT2 cells serving as the primary site of this metabolic shift. Moreover, inhibition of ACAT1 and ACAT2 function significantly exacerbates fibrosis phenotype. Future research must elucidate the mechanism relationship between ACAT1/ACAT2 activity and fibrosis, delineate upstream/downstream effectors of ACAT1/ACAT2, and define the role of metabolically altered AT2 cells in microenvironmental crosstalk. Our work establishes ACAT1/ACAT2 as novel IPF biomarkers and highlights their potential as therapeutic targets to restore metabolic homeostasis in fibrotic lung repair.

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Data Availability

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No datasets were generated during the current study. Primary codes for public datasets analysis were provided at a GitHub repository (https://github.com/KeFang2004/IPF-and-Tryptophan-Metabolism).

Statements and Declarations

Funding

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

Ke Fang (Co-first Author): Conceptualization, Methodology, Software, Investigation, Formal Analysis, Writing - Original Draft;

Xinyue Zhang (Co-first Author): Data Curation, Writing - Original Draft;

Yihan Liu (Co-first Author): Resources, Investigation;

Wangrong Liu: Investigation, Verification;

Yuan Zhang: Funding acquisition, Writing - Review & Editing;

Weijun Cao: Funding acquisition, Writing - Review & Editing.

Ethics Approval

This study was performed in line with the principles of the Declaration of Helsinki. For animal experiments, the approval was granted by the Ethics Committee of Shanghai Pulmonary Hospital

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(approval number 20230809-1). For human sample usage, the approval was granted by the Ethics Committee of Shanghai Pulmonary Hospital (approval number 20240529-1).

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Additional files

Supplementary material 1, Supplementary materials: supplementary figures for the article;

Supplementary material 2, Western blot results: full uncropped images of western blots;

Supplementary material 3, Original data: original data for figure drawing;

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Supplementary material 4, Trp metabolism genes: gene set of tryptophan metabolism used in the study;

Supplementary material 5, Merged expression matrix: expression matrix of the merged data from two bulk RNA-seq data sources;

Supplementary material 6, Differential expression results: differential analysis results of the merged matrix;

Supplementary material 7, WGCNA cyan module genes: genes included in the cyan module in WGCNA results.

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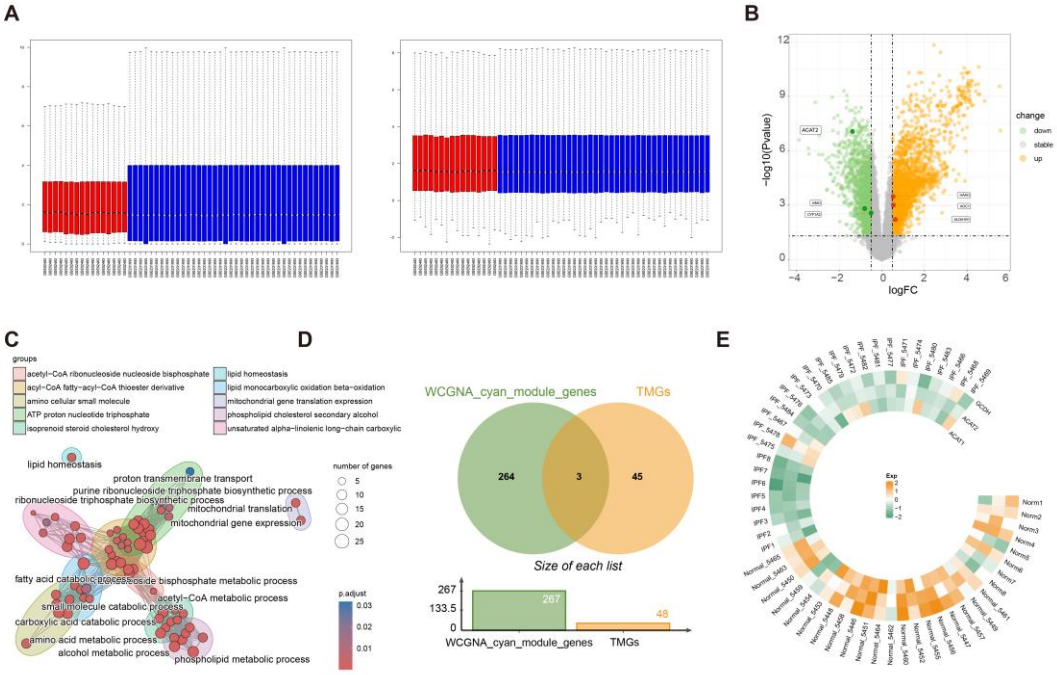

Fig. 1 Bulk RNA-seq analysis for the identification of key TMGs in IPF tissues.

A Batch effect removal between two bulk RNA-seq datasets. **B** Volcano plot showing significant changed TMGs (*ACAT2*, *KMO*, *CYP1A2*, *HAAO*, *AOC1*, *ALDH1B1*). **C** Clustering results of enriched GO terms of WGCNA cyan module genes. **D** Three intersection genes were shared in cyan module genes and TMGs. **E** Heatmap demonstrating the expression level of three key intersection genes (*ACAT1*, *ACAT2*, *GCDH*).

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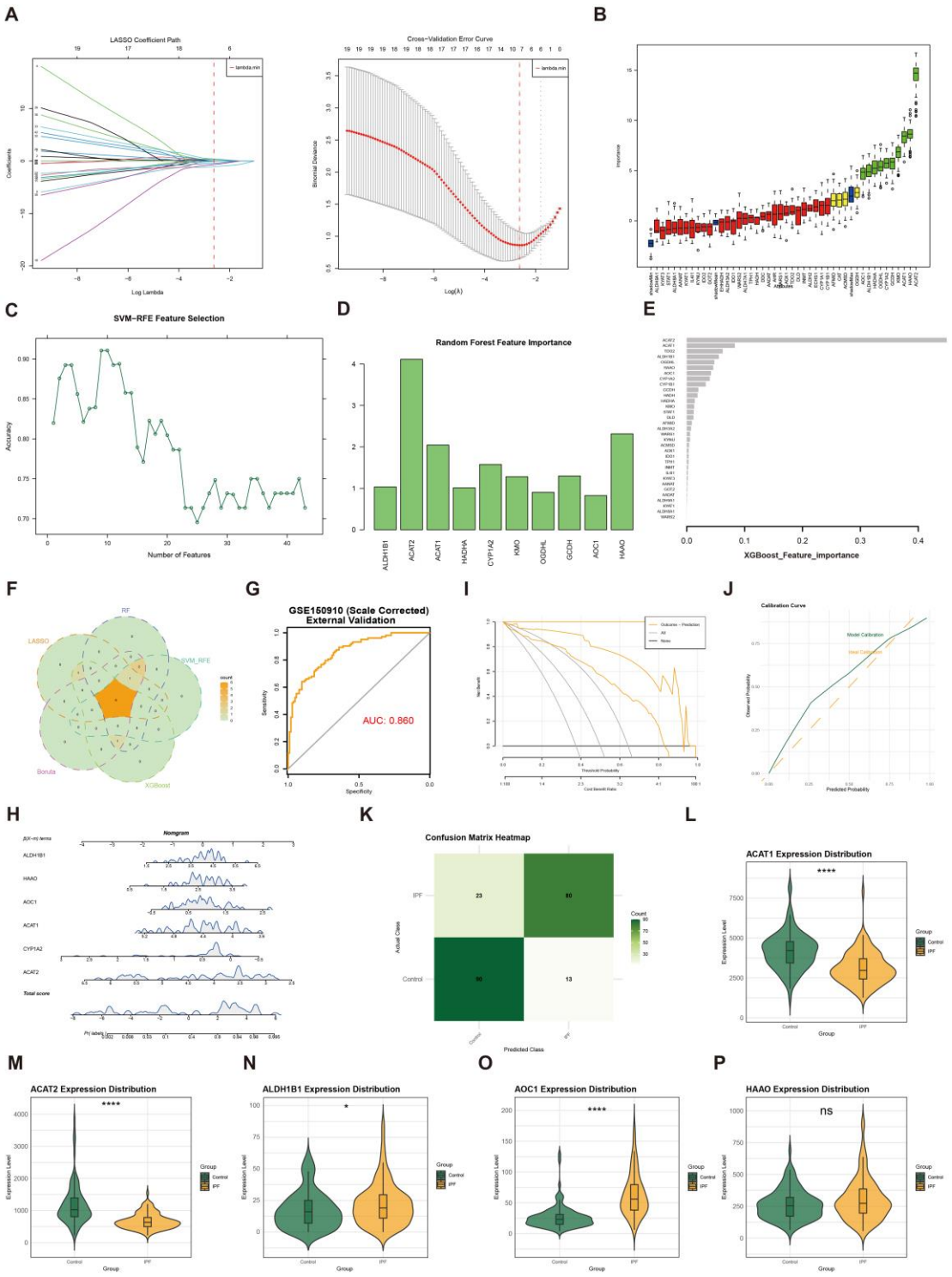

Fig. 2 Combined machine learning screened out key TMGs in IPF.

A-E The LASSO algorithm (**A**), Boruta algorithm (**B**), SVM-RFE algorithm (**C**), random forest (**D**) and XGBoost algorithm (**E**) determined the candidate optimal feature genes. **F** Venn diagram displayed the six optimal key TMGs overlapped by the five above-mentioned machine learning

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outputs. **G** ROC curve of the IPF diagnostic model made with six key TMGs. **H** Nomogram for predicting the likelihood of IPF based on key TMGs expression, providing a visual representation of the robust diagnostic model. **I-J** Decision and calibration curve analysis assessing the nomogram's diagnostic clinical usefulness and accuracy. K Confusion matrix heatmap presenting a high accuracy of the diagnostic model. **L-P** Expression level of key TMGs: *ACAT1* (**L**), *ACAT2* (**M**), *ALDH1B1* (**N**), *AOC1* (**O**), *HAAO* (**P**) displayed by violin plots.

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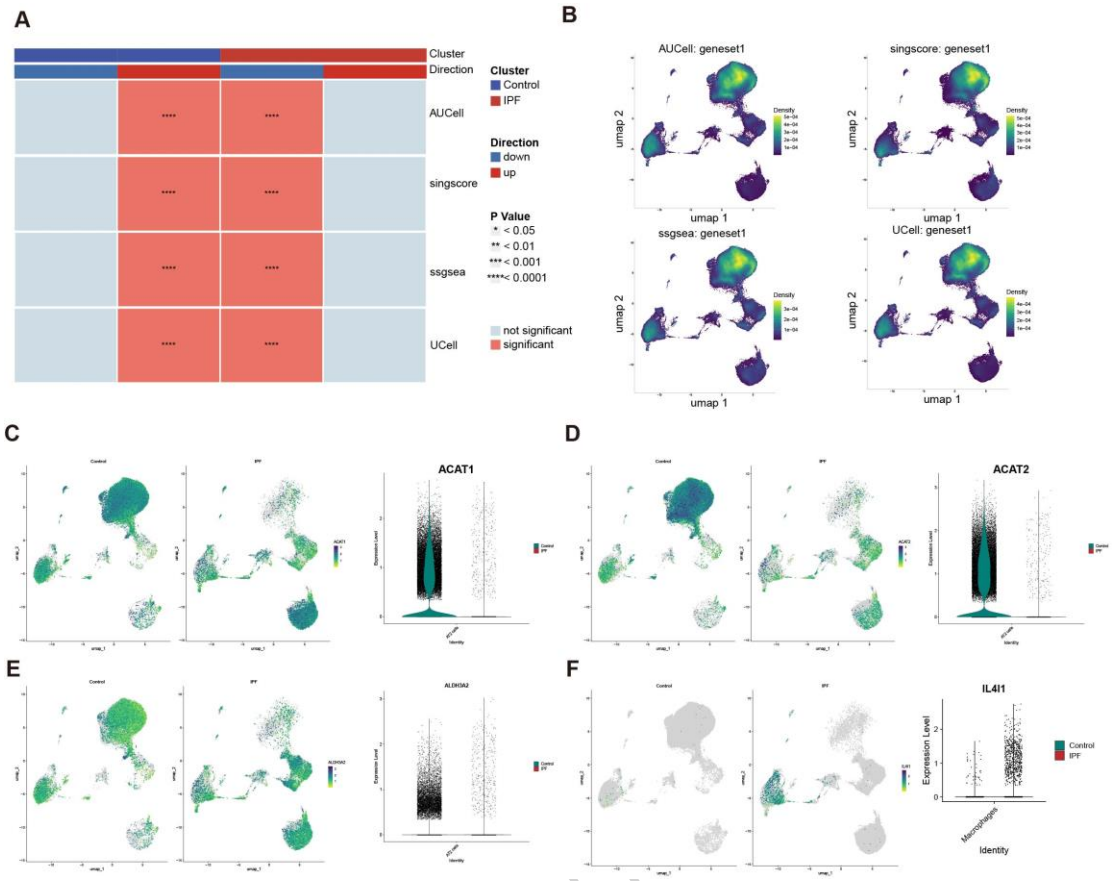

Fig. 3 Tryptophan metabolism scoring among all cell types identified AT2 cells as the main Trp metabolism cells.

A TMGs based gene set scoring showing tryptophan metabolism differences between control and IPF in specific cell type using irGESA R package. **B** Four algorithm calculated scores of TMGs displayed in UMAP, indicating that AT2 cells were the main metabolism functioning cells. **C-E** UMAP plot presents the expression status of three intersection genes (*ACAT1*, *ACAT2* and *ALDH3A2*) between DEGs (IPF vs. control) in AT2 cells and TMGs. **F** UMAP plot presents the expression status of *IL4I1* (the only intersection gene between DEGs (IPF vs. control) in macrophages and TMGs.

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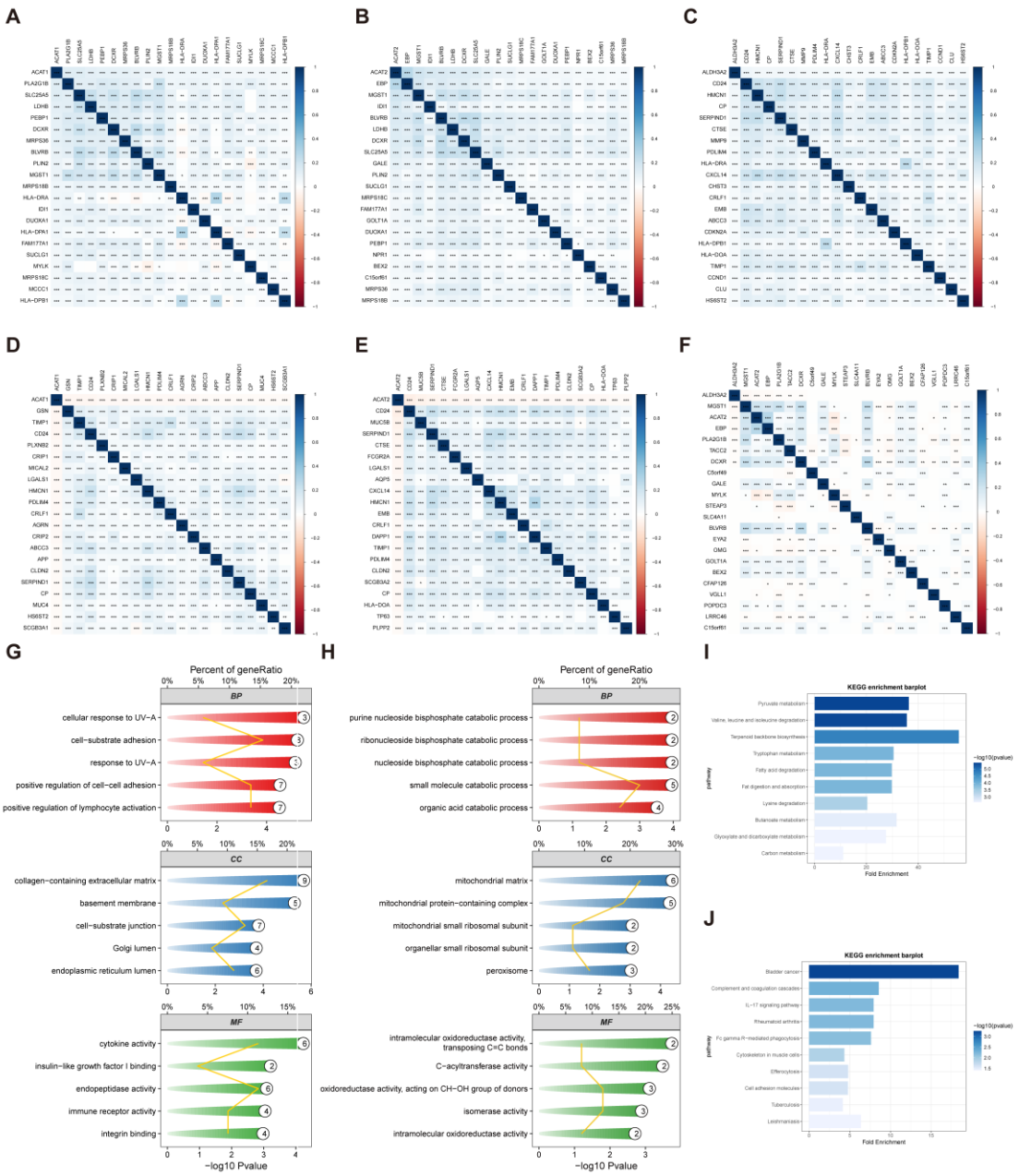

Fig. 4 Correlation and enrichment analysis of three differentially expressed TMGs in AT2 cells.

A-C Heatmaps showing positively correlated genes with *ACAT1* (**A**), *ACAT2* (**B**) and *ALDH3A2* (**C**). **D-F** Heatmaps showing negatively correlated genes with *ACAT1* (**D**), *ACAT2* (**E**) and *ALDH3A2* (**F**). **G** GO enrichment results of union genes of negative correlation genes of the three genes. **H** GO enrichment results of the union genes of the positive correlation genes of the three genes. **I** KEGG enrichment results of union genes of positive correlation genes of the three genes. **J** KEGG enrichment results of union genes of negative correlation genes of the three genes.

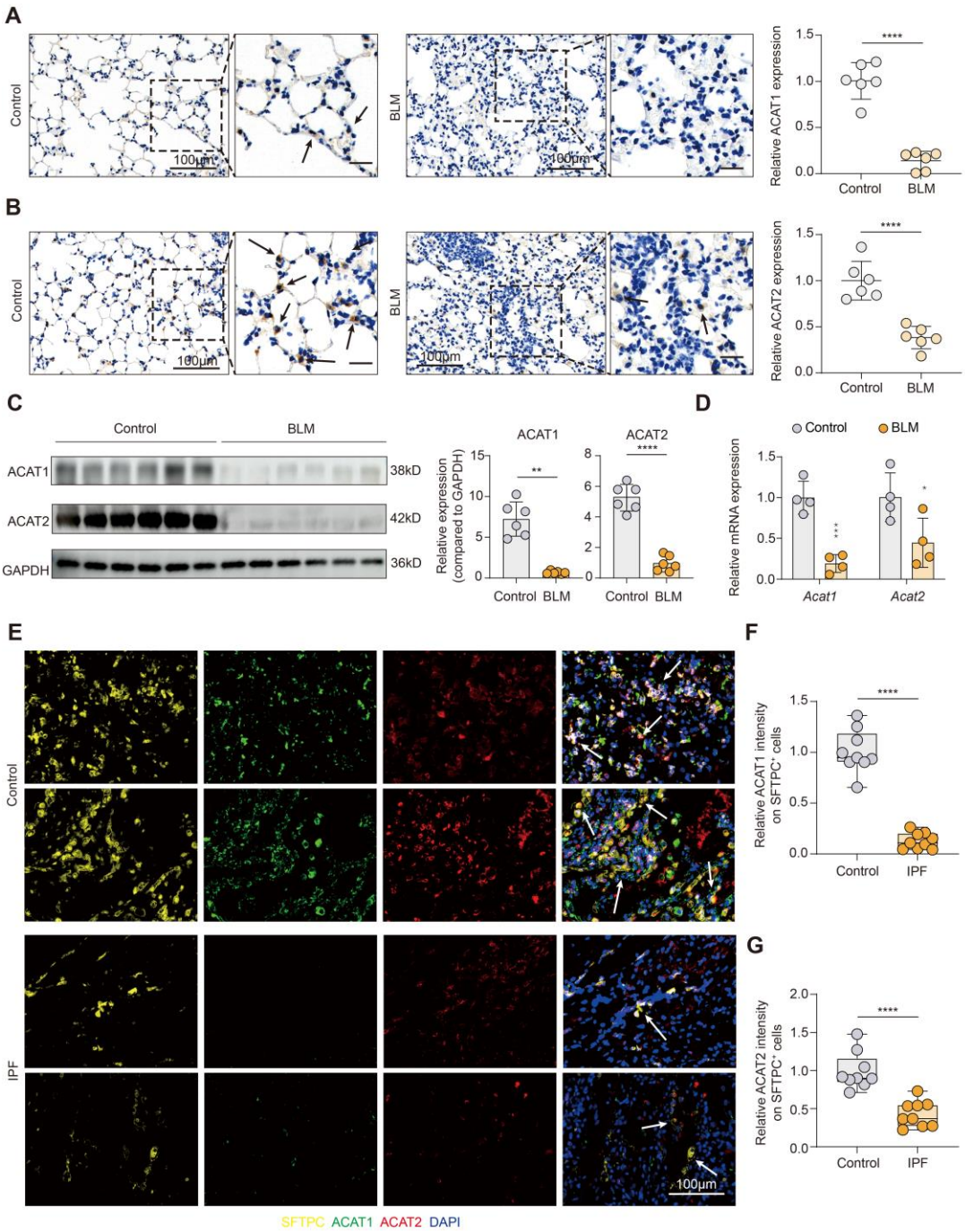

Fig. 5 Biochemical experiments verified down-regulation of ACAT1 and ACAT2 expression in IPF status.

A Immunohistochemical staining against ACAT1 in lung sections and the quantification of the signal strength (n = 6). Scale bar = 100 μm. **B** Immunohistochemical staining against ACAT2 in lung sections and the quantification of the signal strength (n = 6). Scale bar = 100 μm. **C** Western blot results showing down-regulated ACAT1 and ACAT2 protein expression level in IPF mouse

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tissues and bar plots showing the quantification results (n = 6 per group). **D** Relative mRNA expression of *Acat1* and *Acat2* measured by qRT-PCR (n = 4 per group). **E** Immunofluorescence staining of human lung tissues from IPF and non-IPF patients. Scale bar = 100 μm. **F-G** Expression level represented by fluorescence intensity of ACAT1 (**F**) and ACAT2 (**G**) in SFTPC^{+} AT2 cells (n = 3 per group, 3 views per sample). *$p < 0.05$, ***$p < 0.001$, ****$p < 0.0001$ by Student's t tests.

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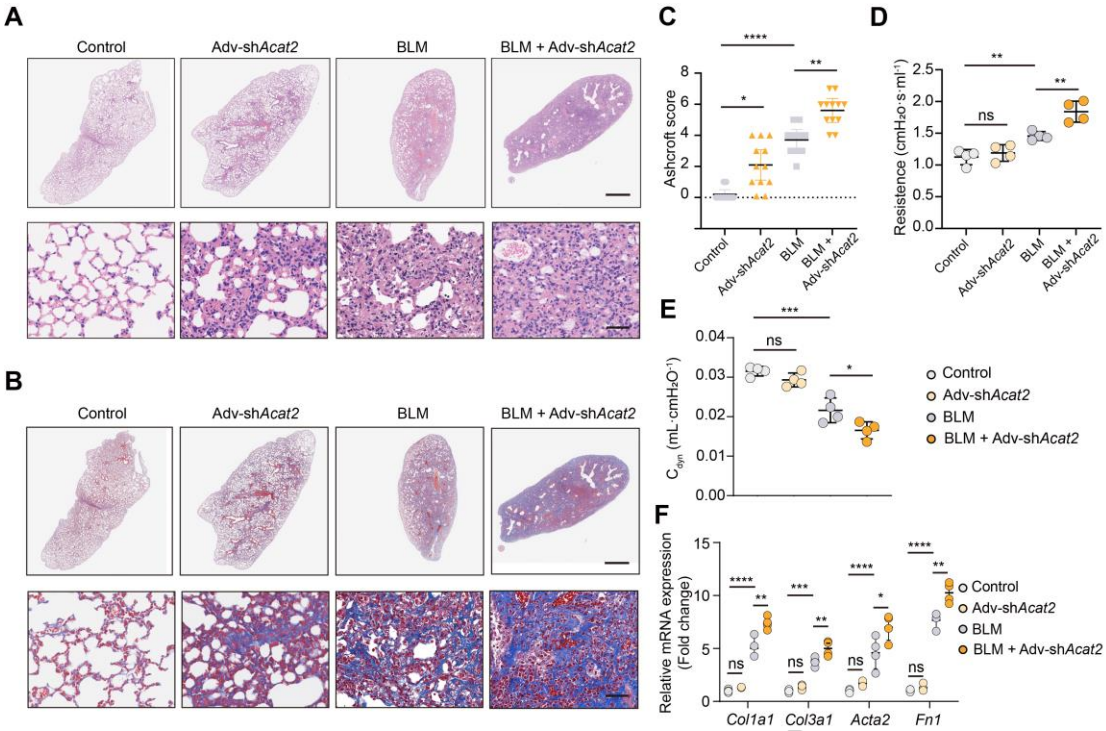

Fig.6 Knockdown of *Acat2* exacerbated pulmonary fibrosis.

A H&E staining of the left lungs from adenovirus-treated or not-treated mice at day 28 post-BLM or -saline administration. Scale bar = 1 mm. Images in the lower panels were magnified from the photomicrographs in the upper panels. Scale bar = 50 μm (n = 4 per group). **B** Masson trichrome staining of left lungs from adenovirus-treated or not-treated mice at day 28 post-BLM or -saline administration. Scale bar = 1 mm. Images in the lower panels were magnified from the photomicrographs in the upper panels. Scale bar = 50 μm (n = 4 per group). **C** Ashcroft score quantified fibrosis level of screen shots from representative lung sections (n = 4 per group, 3 views per slide). **D-E** Lung function parameters including dynamic compliance (C_{dyn}) (**D**) and respiratory resistance (**E**) among different groups were compared 28 days after BLM challenge or adenovirus treatment (n = 4). **F** Fibrosis marker genes relative expression level was shown by RT-PCR (n = 4). Data are presented as the mean ± SEM. ns: not significant, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$ by one-way ANOVA with Tukey's multiple comparison tests (**C**, **D** and **E**) and two-

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way ANOVA with Tukey's multiple comparison tests (**F**).

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